

Research Techniques

Preclinical animal models

Evaluating Animal Behavior

Simple Behavioral Observation

- tremors, ptosis (drooping eyelids), salivation, **cataplexy** (e.g., from motor side effects), etc.

Measures of motor activity

- identify drugs that produce sleep, sedation, loss of coordination, or drugs that stimulate activity
- **open field test**

“Gold Standard” measures of

Learning & memory

Fear / anxiety

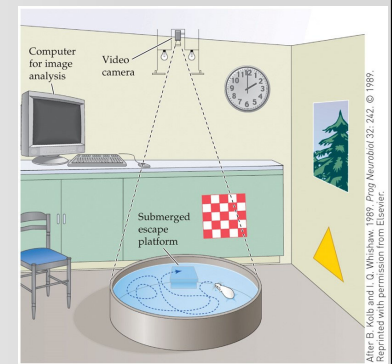
Depression-like behaviors

- behavioral despair / anhedonia

Drugs of abuse

The Morris water maze is a test of spatial learning.

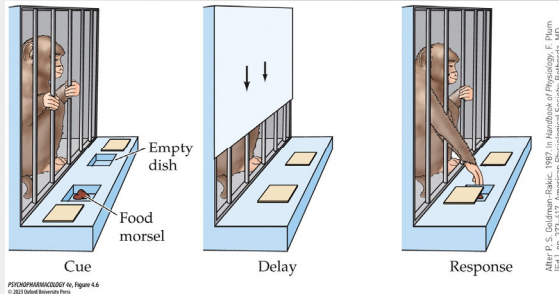
- A circular pool of water made opaque by addition of milk or a dye.
- Animals must swim until they find the escape platform hidden from view just below the water surface.
- The animals learn the spatial position of the platform relative to landmarks outside the pool.



Learning & Memory

The delayed-response test is similar to working memory in humans

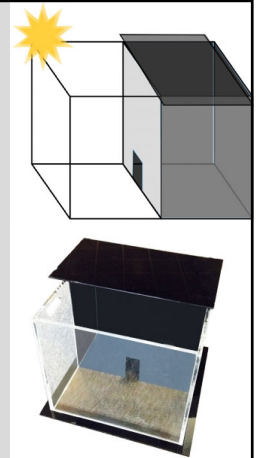
A delay between stimulus presentation (food) and testing evaluates recall for the position of the food



Learning & Memory

The light-dark box measures unconditioned anxiety

- Animals naturally avoid brightly lit places
- L-D box has one side brightly lit and the other side is dark
- Measures include number of crossings, time spent on each side, total motor activity



Measures of Anxiety

The elevated plus-maze measures unconditioned anxiety

- Cross-shaped maze raised off the floor.
- Two arms are open, and two arms have enclosed sides.
- Antianxiety drugs increase the amount of exploration of the open arms; drugs that increase anxiety decrease exploration of open arms.



Measures of Anxiety

Conditioned emotional responses (emotional memories) involve classical (Pavlovian) conditioning

- Presentation of a signal (a light or **tone, CS**) is followed by an unavoidable electric **shock (US)**.
- After conditioning (US+CS pairing), the signal (CS) will elicit a conditioned response (CR or CER)
 - Conditioned freezing
 - Fear-potentiated startle

Fear (Threat) Learning

The forced swim test is a measure of behavioral despair

- Animal must swim in a cylinder from which it can't escape.
- Immobility reflects a lowered mood
- Animal model of hopelessness



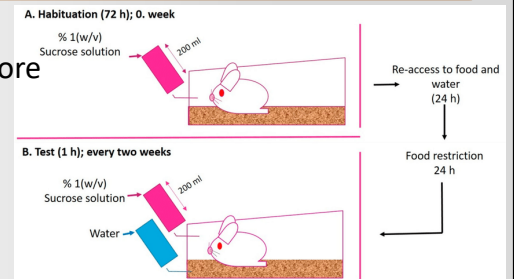
PSYCHOPHARMACOLOGY 4e, Figure 4.8
© 2021 Oxford University Press

Courtesy of Porsolt & Partners Pharmacology

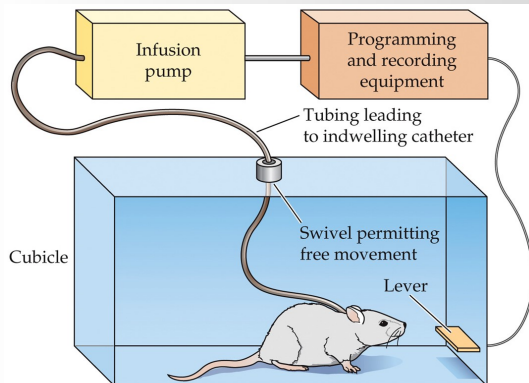
Depressive Behaviors

The sucrose preference test is a measure of anhedonia, the inability to derive pleasure from normally pleasurable events

- Animals are given a choice of water or more palatable sucrose solution.
- A variety of stressors can reduce sucrose preference



Depressive Behaviors



PSYCHOPHARMACOLOGY 4e, Figure 4.9
© 2021 Oxford University Press

Drug self-administration measures the reinforcement value of substances

Drugs of Abuse

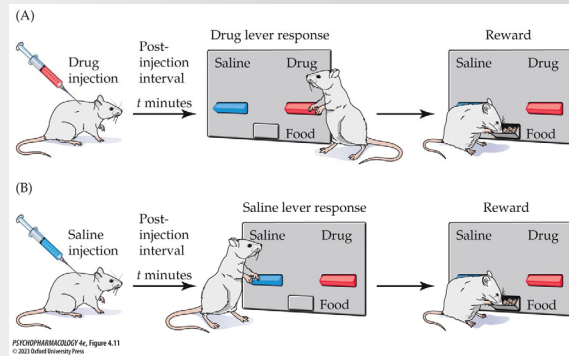
- an indicator of abuse potential in humans
- Varying the schedule of reinforcement determines how hard an animal will work for drug
 - Breaking point: Point at which effort required exceeds reinforcing value—the higher the breaking point, the higher the abuse potential.
- Also used to study withdrawal and/or relapse (reinstatement)

PSYCHOPHARMACOLOGY 4e, Figure 4.9
© 2021 Oxford University Press

Drug self-administration measures the reinforcement value of substances

Drugs of Abuse

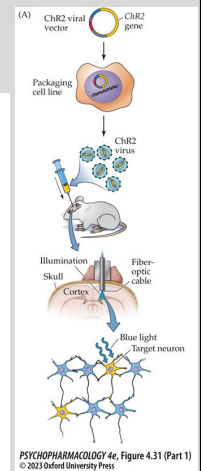
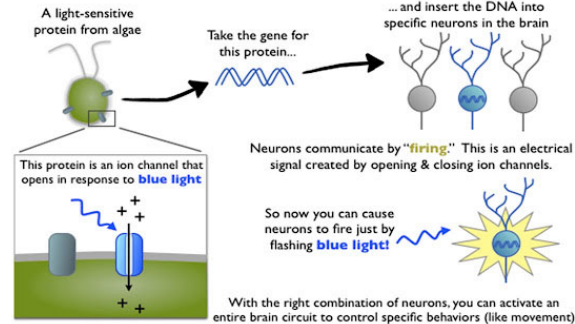
Drug discrimination studies examine how an animal experiences a drug



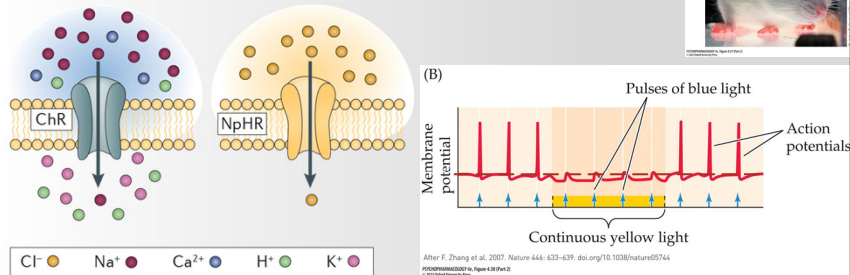
Drugs of Abuse

Optogenetics uses light to activate select populations of neurons

How optogenetics works



Optogenetics



Tye, K. M. & Deisseroth, K. (2012). Optogenetic investigation of neural circuits underlying brain disease in animal models. *Nature Reviews Neuroscience*, 13, 251-266. doi: 10.1038/nrn3171

Chemogenetics uses designer receptors activated by designer drugs (DReADDs) to probe functions of signal transduction systems

